

Notes: Amiti and Weinstein (QJE, 2011)

Exports and Financial Shocks

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1 Big picture

- **Question.** Can deteriorations in bank health explain why exports collapse more than domestic output during financial crises?
- **Core hypothesis.** Exports are more finance-sensitive than domestic sales because international transactions require more working capital and involve more default risk. Banks provide trade finance, so bank distress can transmit directly to exporters.
- **Setting.** The paper studies Japanese listed manufacturing exporters and their main banks during Japanese banking crises from the late 1980s through the late 1990s, with an additional robustness window around the 2008-2009 global financial crisis.
- **Key data advantage.** The authors match individual exporters to the main bank likely to provide their trade finance. This allows them to compare exporters in the same industry-year that happen to rely on healthier versus less healthy banks.
- **Main result.** A decline in the market value of a firm's main bank predicts slower future export growth. The baseline coefficient is about 0.08: when a firm's bank has a 30 percent decline in market-to-book value, the firm's annual exports fall by about 2.7 percent relative to a comparable firm whose bank does not decline.
- **Exports versus domestic sales.** Bank health matters much more for exports than for domestic sales. Among exporters, the bank-health effect on domestic sales is small and statistically insignificant, while the effect on exports relative to domestic sales is large and significant.
- **Mechanism evidence.** The effect is stronger during banking crises, stronger when bank health declines rather than rises, concentrated among sea-shipping industries, and weaker for firms with foreign affiliates. These patterns fit the working-capital and default-risk interpretation.
- **Economic takeaway.** Financial crises reduce trade not only through demand. They also reduce export supply by disrupting the banks that finance exporters.

2 Incremental contribution of the paper

- **From correlation to a bank-firm transmission channel.** Earlier work documented that trade collapses are associated with financial crises or credit conditions. This paper identifies a direct link from a particular bank's health to the export growth of its own client firms.
- **Separating trade finance from generic external finance.** The paper emphasizes that trade finance is not the same as standard external finance dependence. Trade finance covers working capital, payment guarantees, and risk insurance for cross-border transactions.
- **Export-specific effect.** A central contribution is showing that bank health affects exports more than domestic sales within the same firms. This directly addresses why exports fall relative to output during crises.
- **Matched Japanese data.** The paper combines DBJ/Nikkei firm data, Japan Company Handbook bank relationships, bank market-to-book data, and foreign bills bought as a measure of trade finance. This combination is unusually well suited for testing the trade finance channel.

- **Credible identification with rich fixed effects.** Industry-year fixed effects absorb demand, factor prices, exchange rates, and industry-level supply shocks. Bank fixed effects absorb time-invariant differences across banks and their client bases.
- **Mechanism tests.** The paper moves beyond a reduced-form result by showing stronger effects for crisis years, sea-shipped goods, and arm’s-length exporters without foreign affiliates.
- **Magnitude contribution.** The estimated partial effect of financial shocks explains a substantial fraction of crisis-year Japanese export declines: roughly 20–50 percent of the deviation of export growth from trend in some years.

3 Data and measurement

3.1 Exporter data

- The main sample uses listed Japanese manufacturing firms.
- For 1986–1999, the core firm data come from the Development Bank of Japan database of unconsolidated corporate reports.
- For later years, Nikkei NEEDS FinancialQuest provides consolidated export data. The paper uses this later period mainly for robustness checks because bank mergers make bank-firm matching difficult between 2000 and 2006.
- The manufacturing exporters in the sample account for a large share of Japanese merchandise exports, and Figure I shows that aggregate export growth in the sample closely tracks official aggregate export growth.

Interpretation: The sample is not a small corner of the export sector. It captures the export behavior of major Japanese manufacturing exporters.

3.2 Bank-firm matching

- The authors use the *Japan Company Handbook* to identify each firm’s transactional or “reference” banks.
- These banks are listed in order of importance and typically handle payment settlement and foreign exchange transactions.
- If the first listed reference bank is regional and unlikely to provide trade finance, the authors use the first large commercial bank on the list.
- Robustness tests also use alternative definitions: the largest city-bank lender and the first listed reference bank regardless of whether it is a city bank.

Interpretation: The empirical design relies on identifying the bank most likely to supply the exporter’s trade finance. The robustness checks show that the results do not depend on one narrow matching rule.

3.3 Bank health

- The preferred bank-health measure is the log change in the bank's market-to-book value over the 12 months before the close of the firm's accounting year.
- The authors use market values rather than reported capital ratios because Japanese banks in the 1990s had incentives and opportunities to obscure losses.
- Figure II shows substantial cross-bank heterogeneity in changes in market-to-book values, especially during banking stress years.
- A robustness check uses the risk-based capital ratio. The effect remains positive and significant.

Interpretation: The main identifying variation is not simply that Japan had bad banking years. It is that some exporters were tied to banks whose health deteriorated more than others at the same time.

3.4 Trade finance and foreign bills bought

- The paper uses foreign bills bought as an observable component of trade finance extended by banks.
- Foreign bills bought are short-maturity claims related to exporters' financing of trade receivables.
- Table II shows that bank health is more strongly associated with foreign bills bought than with total lending.

Interpretation: A deterioration in bank health seems especially damaging for short-maturity trade finance, which needs continuous renewal.

3.5 Key dependent variables

- The main dependent variable is firm-level export growth, $\Delta \ln(Exports_{ft})$.
- The paper also studies domestic sales growth, $\Delta \ln(DomesticSales_{ft})$.
- A particularly important outcome is the change in exports relative to domestic sales, $\Delta \ln(Exports/DomesticSales)_{ft}$.

Interpretation: The export/domestic-sales comparison is central because it asks whether bank health affects the internationally financed part of firm activity more than the domestic part.

4 Models and empirical specifications

4.1 Bank-level correlations

The paper first documents two reduced-form correlations:

- banks with higher lagged market-to-book values lend more overall and provide more foreign bills bought;
- the exports of firms tied to a bank are positively associated with that bank's trade finance.

Conceptually, the bank-level specification relates bank lending or foreign bills bought to lagged bank health with bank and year fixed effects:

$$\ln(Lending_{bt}) = \alpha_b + \delta_t + \beta \ln(MTB_{b,t-1}) + \varepsilon_{bt}.$$

Interpretation: This stage motivates the causal analysis. It shows that trade finance is linked to bank health and to client exports, but by itself it does not prove causality.

4.2 Baseline firm-level export regression

The main specification is:

$$\Delta \ln(Exports_{ft}) = \sum_{i,t} \alpha_{it} IND_{fit} + \sum_b \gamma_b BANK_{fb,t-1} + \beta \sum_b \Delta \ln(MTB_{bt-1}) BANK_{fb,t-1} + \varepsilon_{fit}.$$

- IND_{fit} are industry-by-year indicators.
- $BANK_{fb,t-1}$ indicates the firm-bank relationship.
- $\Delta \ln(MTB_{bt-1})$ is the lagged change in the bank's market-to-book value.
- β is the main coefficient: it measures whether exporters tied to healthier banks have higher subsequent export growth.

Interpretation: Identification comes from comparing firms in the same industry-year but with different main banks.

4.3 Export-specificity specification

To test whether the effect is specific to exports, the authors replace the dependent variable with domestic sales growth and with export growth relative to domestic sales:

$$\Delta \ln \left(\frac{Exports}{DomesticSales} \right)_{ft}.$$

Interpretation: If bank health mattered only through generic firm demand or broad firm distress, it should affect domestic sales similarly. Instead, the export/domestic-sales result points to trade finance.

4.4 Endogeneity checks

The paper addresses reverse causality and omitted variables through several approaches:

- adding lagged export growth in an Arellano-Bond specification;
- adding lagged firm assets and profitability;
- restricting the sample to pre-Asian-crisis years;
- controlling for destination-weighted foreign GDP growth;
- using an IV based on the residual of bank health after removing the component correlated with the firm's own share price.

Interpretation: These tests are designed to rule out the concern that firm export prospects caused bank health changes rather than the reverse.

4.5 Heterogeneity specifications

The paper estimates interactions of bank health with variables that should matter if the mechanism is trade finance:

- crisis-year dummies;
- indicators for bank market-to-book declines;
- air-shipping industries;
- foreign-affiliate status.

Interpretation: The results are strongest exactly where the trade finance channel predicts they should be strongest: crisis years, declining banks, slower sea-shipped goods, and arm's-length transactions with default risk.

5 Identification and empirical design

5.1 What the design can identify

- The design identifies whether changes in a bank's health predict export growth for client firms relative to other firms in the same industry-year.
- Industry-year fixed effects remove common shocks to each industry at each date, including demand conditions and exchange-rate-related industry shocks.
- Bank fixed effects remove stable differences across banks, such as the possibility that some banks serve systematically higher-growth exporters.

Interpretation: The coefficient is not just a macro crisis correlation. It is a within-industry-year comparison across exporters with different bank exposure.

5.2 Main identifying assumption

The central identifying assumption is that, conditional on industry-year and bank fixed effects, changes in a bank's market-to-book value are not driven by future export shocks of a particular

client firm.

This assumption is plausible for two reasons:

- an individual exporter is tiny relative to the bank’s total loan portfolio;
- Japanese banking crises were driven largely by banking and real estate shocks, not by changes in the export prospects of individual manufacturing firms.

5.3 Threats to identification

- **Reverse causality.** Strong or weak exporters could affect bank valuations. The paper addresses this with an IV that purges bank health of firm share-price movements.
- **Firm-bank selection.** Good exporters could match with good banks. Bank fixed effects and the no-bank-switcher sample address this concern.
- **Industry or destination demand shocks.** Industry-year fixed effects and destination GDP controls address common demand shocks.
- **Export entry and exit.** Heckman and Tobit specifications show that the core result is not driven by selection into or out of exporting.

5.4 What remains a limitation

- The paper observes a major component of trade finance, foreign bills bought, but not every possible trade-finance instrument.
- The bank-firm match identifies the likely trade-finance bank, but firms may have multiple financial relationships.
- The estimated coefficient is a partial-equilibrium effect of bank health on client exporters, not a full general-equilibrium decomposition of the entire trade collapse.

Interpretation: The evidence is strongest for the existence and size of a bank-to-exporter channel, not for ruling out every other driver of trade collapses.

6 Tables and figures

6.1 Figure I: Firm-level and aggregate export growth

Read:

- Figure I compares official Japanese aggregate export growth with aggregate export growth computed from the authors’ firm-level data.
- The two series move closely together during the 1987–2009 period.
- The figure shows sharp export contractions around Japanese banking stress episodes and the 2008–2009 global financial crisis.

Economic message: The firm-level sample is representative enough to study movements in Japanese exports. This supports the paper’s use of listed manufacturing exporters to analyze aggregate export dynamics.

6.2 Figure II: Change in market-to-book value of Japanese banks

Read:

- Figure II plots changes in market-to-book values across Japanese main banks over time.
- The median bank line falls sharply in crisis periods, but the dots show large dispersion across banks.
- This dispersion is especially important for identification because different firms are linked to different banks.

Economic message: The figure provides the main source of treatment variation. Even in the same macro crisis, some banks become much less healthy than others.

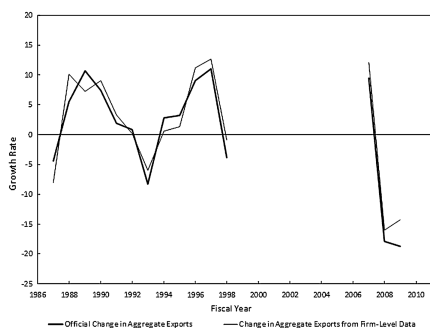


FIGURE I
Firm-Level and Aggregate Export Growth in Japan, 1987–2009
Source: Firm-level data is from the Development Bank of Japan (2004), “Corporate Finance Data Bank.” The aggregate official exports for each fiscal year was downloaded from the Japanese Ministry of Finance (<http://www.customs.go.jp/toukei/sui/html/time.e.htm>).

(a) Figure I: Firm-level and aggregate export growth.

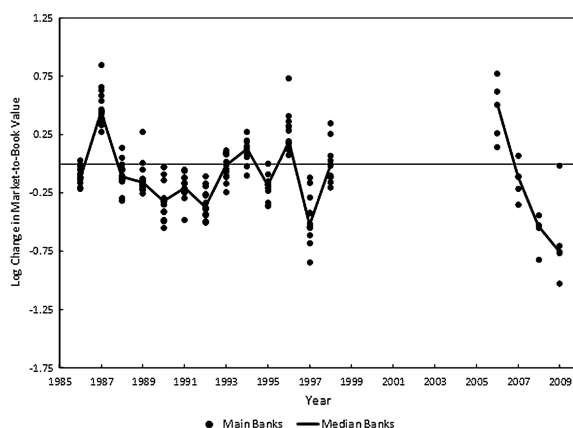


FIGURE II
Change in the Market-to-Book Value of Japanese Banks, 1986–2009
Source: Development Bank of Japan (2004), “Corporate Finance Data Bank.”

(b) Figure II: Bank market-to-book changes.

6.3 Table I: Summary statistics

Read:

- Table I reports statistics for banks and firms.
- Trade finance is small relative to total loans: foreign bills are less than 1 percent of typical bank lending.
- Firm-level data include export growth, domestic sales growth, export-to-domestic-sales growth, firm balance-sheet variables, and bank exposure variables.
- In the 2008–2010 sample, average export growth is negative and bank market-to-book changes are strongly negative.

Economic message: Banks are not being mechanically driven by the fate of a single exporter. Exporter exposure is tiny relative to bank balance sheets, which supports the direction from bank health to exporter outcomes.

6.4 Table II: Bank health, trade finance, and exports

Read:

- Columns 1–3 show that healthier banks lend more and provide more trade finance.
- The coefficient of bank health on foreign bills bought is larger than on total loans.
- Columns 4–5 show that bank-level trade finance is positively associated with client exporters’ export growth.

Economic message: Trade finance is especially sensitive to bank health, and client exports move with bank-provided trade finance. This table motivates the firm-level causal analysis.

TABLE I
SUMMARY STATISTICS

By banks (1987–1999)	<i>N</i>	Mean	Median	sd	Min	Max
$\ln(\text{foreign bills bought})_{b,t}$	170	12.06	12.30	0.84	10.13	13.66
$\Delta \ln(\text{foreign bills bought})_{b,t}$	170	−0.05	−0.06	0.19	−0.55	0.48
$\ln(\text{totloans})_{b,t}$	170	16.76	16.84	0.55	15.38	17.56
$(\text{foreign bills}/\text{totloans})_{b,t}$	170	0.01	0.01	0.01	0.00	0.08
$\ln(\text{foreign bills}/\text{totloans})_{b,t}$	170	−4.70	−4.70	0.72	−6.45	−2.56
$\ln(\text{market-to-book value})_{b,t-1}$	170	1.33	1.35	0.52	−0.16	2.61
$\Delta \ln(\text{market-to-book value})_{b,t-1}$	170	−0.07	−0.12	0.28	−0.84	0.84
By firms (1987–1999)						
Number of firms _{<i>t</i>}	7, 173	632	636	46	539	716
$\Delta \ln(\text{exports})_t$	7, 173	0.01	0.03	0.26	−1.23	0.95
$\Delta \ln(\text{domestic sales})_t$	7, 050	0.01	0.01	0.11	−0.43	0.37
$\Delta \ln(\text{exports}/\text{domestic sales})_t$	7, 050	0.00	0.01	0.26	−1.35	1.18
$\Delta \ln(\text{market-to-value})_{f,t-1}$	7, 173	−0.07	−0.12	0.29	−1.26	1.11
$\Delta \ln(\text{assets})_{f,t-1}$	6, 909	0.04	0.02	0.11	−1.03	0.88
$\Delta(\text{profits})_{f,t-1}$	6, 909	0.00	0.00	0.06	−1.77	2.23
$\Delta \ln(\text{share price})_{f,t-1}$	6, 902	−0.03	−0.02	0.30	−3.06	1.55
Total loan _{<i>bt</i>} /Total loan _{<i>ft</i>}	5, 723	0.0001	0.00004	0.0003	0	0.006
By firms (2008–2010)						
$\Delta \ln(\text{exports})_t$	1, 619	−0.08	−0.09	0.23	−0.94	0.70
$\Delta \ln(\text{market-to-value})_{f,t-1}$	1, 619	−0.53	−0.53	0.27	−1.03	0.07

Notes: Profits are defined as the ratio of after-tax net income to total assets.

(a) Table I: Summary statistics.

TABLE II
ASSOCIATIONS BETWEEN BANK HEALTH, TRADE FINANCE, AND EXPORTS

Dependent variable	$\ln(\text{totloans})_{b,t}$	$\ln(\text{foreign bills bought})_{b,t}$	$\ln(\text{bills}/\text{totloans})_{b,t}$	$\Delta \ln(\text{exports})_{b,t}$	$\Delta \ln(\text{exports})_{b,t}$
	(1)	(2)	(3)	(4)	(5)
				By bank	By bank/ industry
$\ln(\text{market-to-book value})_{b,t-1}$	0.144*** (0.054)	0.456*** (0.142)	0.311** (0.137)		
$\Delta \ln(\text{trade finance loan})_{b,t}$				0.155*** (0.062)	0.073*** (0.020)
Fixed effects					
Year	yes	yes	yes	yes	yes
Bank	yes	yes	yes	yes	yes
Industry	no	no	no	no	yes
Observations	185	185	185	138	2,336
Adjusted <i>R</i> ²	0.95	0.92	0.91	0.19	0.10

Notes: Notes: Robust standard errors are in parentheses. *** Significant at the 1% level. ** Significant at the 5% level. Column (1)–(3) include the 15-city banks in our sample: Asahi Bank, Bank of Tokyo, Bank of Tokyo-Mitsubishi, Dai-Itte Kangyo Bank, Daiwa Bank, Fuyo Bank, Hokkaido Tokai-Mitsubishi Bank, Industrial Bank of Japan, Long Term Credit Bank of Japan, Sanjima Bank, Sakura Bank, Sanwa Bank, Sumitomo Bank, Teikoku Bank, Tokai Bank. Column (4) aggregates firm-level exports by first reference bank. Column (5) aggregates firm-level exports by first reference bank within an industry. In columns (4) and (5), we drop the first year that a bank merges to avoid big jumps in exports due to mergers; we drop any firm that withdraws their main bank, and we keep the sample balanced so it only includes firms that were in the sample for the whole sample period.

(b) Table II: Bank health, trade finance, and exports.

6.5 Table III: Exports and trade finance

Read:

- Columns 1–3 estimate the baseline export-growth regression with increasingly rich fixed effects.
- The core coefficient remains close to 0.08 with industry-year and bank fixed effects.
- Column 4 shows that the effect is concentrated in the one-year lag of bank health, not in contemporaneous or two-year lagged bank health.
- Columns 5–7 compare domestic sales and exports relative to domestic sales.

Economic message: Deteriorating bank health causes lower future export growth. The fact that the same bank-health shock has little effect on domestic sales among exporters is the key evidence that this is an export-finance channel rather than a generic firm-performance channel.

6.6 Table IV: Endogeneity

Read:

- Table IV tests whether the baseline result survives major endogeneity concerns.
- The coefficient is stable when adding lagged export growth, lagged firm assets and profits, and restricting to the pre-Asian-crisis period.
- Destination-weighted GDP growth predicts exports, but including it does not remove the bank-health effect.
- The IV specifications use bank-health residuals adjusted for firm share-price movements and still produce similar estimates.

Economic message: The main result is not plausibly driven by exporters' own performance affecting their banks. The effect remains after controlling for firm health, destination demand, and the firm-share-price component of bank health.

Dependent variable	$\Delta \ln(\text{exports})_{j,t}$			$\Delta \ln(\text{domestic sales})_{j,t}$			$\Delta \ln(\text{exports}/\text{domestic sales})_{j,t}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
				MTB lag structure	Non-exporters	Exporters	Exporters
$\Delta \ln(\text{market-to-book value})_{j,t}$				-0.024 (0.018)			
$\Delta \ln(\text{market-to-book value})_{j,t-1}$	0.080*** (0.014)	0.084*** (0.018)	0.082*** (0.018)	0.074*** (0.017)	0.022*** (0.007)	0.005 (0.007)	0.072*** (0.020)
$\Delta \ln(\text{market-to-book value})_{j,t-2}$				-0.016 (0.025)			
Fixed effects							
Year-industry	no	yes	yes	yes	yes	yes	yes
Year	yes	no	no	no	no	no	no
Bank	no	no	yes	yes	yes	yes	yes
Observations	7,173	7,173	7,173	7,114	5,506	7,050	7,050
Adjusted R ²	0.07	0.15	0.15	0.15	0.40	0.42	0.13

Notes. Robust standard errors corrected for clustering at the bank level are in parentheses. ***Significant at the 1% level. **Significant at the 5% level. Estimation is for the period 1987 to 1999. There are 108 industry codes in manufacturing corresponding to the 1982 industry classification. In column (3) and all subsequent columns, the industry dummies are multiplied by each year dummy. The sample in column (5) includes all observations where exports in period t or $t-1$ were 0. In columns (6) and (7), the sample includes all

(a) Table III: Exports and trade finance.

Dependent variable	$\Delta \ln(\text{exports})_{j,t}$				$\Delta \ln(\text{exports}/\text{domestic sales})_{j,t}$	
	(1)	(2)	(3)	(4)	(5)	(6)
	With lagged dependent variable (Arellano-Bond)	With lagged firms' profits	Pre-Asian crisis years (1987 to 1996)	With destination GDP growth	IV Instrument adjusted for firms' share prices	IV Instrument adjusted for firms' share prices
$\Delta \ln(\text{market-to-book value})_{j,t-1}$	0.084*** (0.021)	0.082*** (0.019)	0.079*** (0.026)	0.091** (0.037)	0.078*** (0.017)	0.088*** (0.023)
$\Delta \ln(\text{exports})_{j,t-1}$	-0.093*** (0.022)					
$\Delta \ln(\text{assets})_{j,t-1}$		0.223*** (0.032)				
$\Delta \ln(\text{profits})_{j,t-1}$		-0.006 (0.059)				
$\Delta \ln(\text{wGDP})_{j,t}$				0.019** (0.008)		
$\Delta \ln(\text{wGDP})_{j,t-1}$				0.021** (0.008)		
Fixed effects						
Year-industry	yes	yes	yes	yes	yes	yes
Bank	yes	yes	yes	yes	yes	yes

(b) Table IV: Endogeneity, part 1.

6.7 Table V: Alternative measures of main bank and market timing

Read:

- Table V changes the bank-firm matching method and the bank-health measure.
- The effect remains when the main bank is defined as the largest city-bank lender.
- The effect remains when the first listed reference bank is used even if it is not a city bank.
- It remains when market-to-book is averaged over the last three months, when only March accounting periods are used, and when risk-based capital ratios replace market-to-book values.

Economic message: The result is not an artifact of a single bank-matching rule, a particular month of market valuation, or the preferred bank-health measure.

TABLE IV (CONTINUED)					
Dependent variable:	$\Delta \ln(\text{exports})_{j,t}$			$\Delta \ln(\text{exports}/\text{domestic sales})_{j,t}$	
	(1)	(2)	(3)	(4)	(5)
	With lagged dependent variable (Arellano–Bond)	With lagged firms' profits	Pre-Asian crisis years (1987 to 1996)	With destination GDP growth	IV Instrument adjusted for firms' share prices
First-stage:					
$\Delta \ln(\text{adjusted market-to-book value})_{j,t-1}$					1.000*** (0.001)
$F\text{-stat}$					1.6e+06 (0.001)
Observations	5,925	6,309	5,417	985	6,902
Adjusted R^2		0.16	0.16	0.17	6,902

Notes: Robust standard errors corrected for clustering at the bank level are in parentheses. ***Significant at the 1% level. **Significant at the 5% level. *Significant at the 10% level. Profits are defined as the ratio of after-tax net income to total assets. The variable $\Delta \ln(\text{GDP})$ is the weighted foreign GDP growth where the weights are given by the firm's export share to that destination. Exports by destination are only available for 1988 and 1989, so we use the average of those two years for the period 1987 to 1999. The instrument in columns (5) and (6) is the residual from a regression of the change in the bank's market-to-book value on the change in the firm's share price.

(a) Table IV: Endogeneity, part 2.

TABLE V ALTERNATIVE MEASURES OF MAIN BANK AND MARKET TIMING					
Dependent variable: $\Delta \ln(\text{exports})_{j,t}$					
	(1)	(2)	(3)	(4)	(5)
	Alternative bank matching Largest lender	Ref banks	MTB value: 3 months average	March accounting period	Alternative bank health measure
$\Delta \ln(\text{market-to-book value})_{j,t-1}$	0.074*** (0.018)	0.065*** (0.015)	0.060*** (0.021)	0.063*** (0.020)	
$\Delta \ln(\text{risk-based capital ratio})_{j,t-1}$					0.201** (0.096)
Fixed effects					
Year–industry	yes	yes	yes	yes	yes
Bank	yes	yes	yes	yes	yes
Observations	7,090	7,049	7,089	6,021	2,905
Adjusted R^2	0.13	0.15	0.15	0.20	0.13

Notes: Robust standard errors corrected for clustering at the bank level are in parentheses. ***Significant at the 1% level. **Significant at the 5% level. In column (1), we use an alternative method for matching firms to banks: we assign a city bank that was the largest loan provider that year. If the exporter had no loans from a city bank that year, we assign the previous year's city bank. In column (2), we use the first listed reference bank from the company handbooks, even if the first reference bank is not a city bank. In column (3), we define the market-to-book value as the average of the last three months of the accounting period. In column (4), we only keep observations where the accounting period ended in March. In column (5), we measure bank health using the risk-based capital ratio instead of the market-to book ratio.

(b) Table V: Alternative bank and timing measures.

6.8 Table VI: Heterogeneous effects

Read:

- Column 1 shows that the bank-health effect is much stronger in Japanese banking crisis years.
- Column 2 shows a similar effect in the 2008–2010 global crisis period.
- Columns 3–4 show that bank-health declines matter more than bank-health improvements.
- Column 5 shows that bank health matters for industries whose exports are mainly shipped by sea, but not for industries whose exports are mainly shipped by air.
- Column 6 shows weaker effects for firms with foreign affiliates.

Economic message: This is the mechanism table. The effect is strongest where firms need bank-provided working capital and default-risk protection the most.

TABLE VI HETEROGENEOUS EFFECTS						
Dependent variable: $\Delta \ln(\text{exports})_{j,t}$	Crisis interaction	Subsample 2008–2010	Asymmetric effects	Non-linearities	With air interaction	With foreign interaction
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \ln(\text{MTB})_{j,t-1}$	0.038 (0.027)	0.091*** (0.034)	0.030 (0.041)	0.063** (0.030)	0.120*** (0.022)	0.135*** (0.025)
Crisis* $\Delta \ln(\text{market-to-book value})_{j,t-1}$	0.110*** (0.036)					
Down* $\Delta \ln(\text{MTB})_{j,t-1}$			0.079* (0.044)			
Up.q1* $\Delta \ln(\text{MTB})_{j,t-1}$				–0.032 (0.059)		
Down.q4* $\Delta \ln(\text{MTB})_{j,t-1}$				0.046 (0.036)		
Air* $\Delta \ln(\text{MTB})_{j,t-1}$					–0.130*** (0.037)	
Foreign affiliate* $\Delta \ln(\text{MTB})_{j,t-1}$						–0.077*** (0.019)
Foreign affiliate ₇						0.024 (0.015)

(a) Table VI: Heterogeneous effects, part 1.

TABLE VI (CONTINUED)						
Dependent variable: $\Delta \ln(\text{exports})_{j,t}$	Crisis interaction	Subsample 2008–2010	Asymmetric effects	Non-linearities	With air interaction	With foreign interaction
	(1)	(2)	(3)	(4)	(5)	(6)
Joint significance tests						
H0: $\beta_{\Delta \ln(\text{MTB})} = 0$					–0.010 (0.038)	
H0: $\beta_{\Delta \ln(\text{MTB})} = 0$						0.058* (0.029)
$F\text{-test: } \beta_{\text{Up,q1}} + \beta_{\text{Down,q4}} = 0$				3.36* (0.078)		
Fixed effects						
Year–industry	yes	yes	yes	yes	yes	yes
Bank	yes	yes	yes	yes	yes	yes
Observations	7,173	1,619	7,173	7,173	4,933	1,756
Adjusted R^2	0.15	0.19	0.15	0.15	0.14	0.14

Notes: MTB is "market-to-book value." Robust standard errors corrected for clustering at the bank level are in parentheses. ***Significant at the 1% level. **Significant at the 5% level. *Significant at the 10% level. Crisis dummy equals 1 for years 1990, 1995, 1997. In column (1), down is defined as equal to 1 if the market-to-book value falls. In column (4), up.q1 is defined as the top quartile of the change in the market-to-book value, and down.q4 is the bottom quartile. In column (5), the air dummy is defined as equal to 1 if the share of trade exported by air is greater than 50% within that industry. Miscellaneous industries were dropped because of convergence difficulties. The foreign affiliate data was only available for 1998 and 1999. Foreign affiliate is equal to 1 if the firm reported nonzero foreign assets during those years, which we apply to the sample period 1997 to 1999.

(b) Table VI: Heterogeneous effects, part 2.

6.9 Table VII: Selection

Read:

- Column 1 drops bank switchers, and the result remains.
- Column 2 uses the percentage change in market-to-book, which accommodates bank failures better than log changes.
- Column 3 uses a Heckman selection correction for export participation.
- Columns 4–5 use Tobit specifications that allow firms entering or exiting export markets to remain in the analysis.

Economic message: Selection into banks or into exporting does not explain the main result. The result appears to reflect changes in export intensity among firms tied to distressed banks.

TABLE VII SELECTION					
Dependent variable	$\Delta \ln(\text{exports})_{f,t}$			Percentage change in exports	Change in exports/total sales
	No bank switchers	Percentage change in MTB value	Heckman selection	Tobit (random effects)	Tobit (random effects)
	(1)	(2)	(3)	(4)	(5)
$\Delta \ln(\text{market-to-book value})_{f,t-1}$	0.084*** (0.018)	0.065*** (0.019)	0.086*** (0.020)	0.079*** (0.021)	0.008*** (0.003)
Inverse mills ratio			−0.756 (0.565)		
Fixed effects					
Year–industry	yes	yes	yes	no	no
Bank	yes	yes	yes	yes	yes
Year	no	no	no	yes	yes
First stage					
Relative value added per worker $_{f,t-1}$			0.384*** (0.070)		
$\Delta \ln(\text{market-to-book value})_{f-1}$			−0.056 (0.108)		
Fixed effects:					
Year			yes		
Bank			yes		
Observations	6,616	7,179	8,380	7,277	8,250
R-squared	0.16	0.15			

Table VII: Selection.

7 Short summary

- The paper studies whether bank distress reduces exports more than domestic sales.
- It uses Japanese bank-firm matched data that identify each exporter’s likely trade-finance bank.
- Bank health is measured mainly by changes in market-to-book value.

- The baseline coefficient implies that a 30 percent decline in bank health reduces client-firm export growth by about 2.7 percent.
- The effect is much smaller for domestic sales and significant for exports relative to domestic sales.
- Endogeneity tests show that the result is not driven by export demand, firm performance, or firm share prices.
- Robustness tests show that the result does not depend on the bank matching rule or the bank-health measure.
- Mechanism tests show larger effects in crisis years, for sea-shipped goods, and for firms without foreign affiliates.
- The estimated partial effect is economically large, accounting for a substantial fraction of crisis-year Japanese export declines.

8 Conclusion

8.1 Core conclusion

Amiti and Weinstein show that financial shocks to banks have large effects on the exports of their client firms. The evidence supports a trade-finance channel: when exporters' banks become less healthy, the exporters' foreign sales fall relative to domestic sales. This provides a micro-level explanation for why exports can collapse more than output during financial crises.

8.2 Economic intuition

Exporting differs from domestic selling in two key ways. First, international transactions take longer to complete, especially when goods are shipped by sea, so exporters need working capital for a longer time. Second, international transactions expose exporters to foreign buyer default risk, so they rely on banks for guarantees and payment insurance. When banks become unhealthy, the supply of this finance contracts. Exporters tied to troubled banks therefore reduce exports even if their underlying products and domestic sales remain relatively intact.

8.3 Incremental contribution revisited

The paper advances the literature in three main ways. First, it directly matches exporters to the banks that finance them. Second, it separates export effects from domestic sales effects within firms. Third, it uses heterogeneity across crisis years, shipping mode, and foreign-affiliate status to show that the results are consistent with trade finance rather than generic credit constraints.

8.4 Implications

The findings imply that trade finance should be central in analyses of financial crises and trade collapses. Stabilizing banks may support exports not only by improving macro conditions but also

by preserving the short-term financial infrastructure that exporters need. The paper also suggests that countries with export sectors heavily dependent on external trade finance may be especially vulnerable when domestic banks are impaired.

8.5 Limitations

The paper studies listed Japanese manufacturing firms, so external validity to small firms or other countries is not automatic. The empirical design identifies the bank-client channel but does not directly observe every trade-finance contract. The estimated effects are partial effects and do not fully decompose the entire aggregate trade collapse. Still, the evidence is strong that the financial sector is a major part of the export-collapse mechanism.

8.6 Takeaway

The enduring takeaway is that exports are finance-intensive in a way that domestic sales are not. Banks are not just passive intermediaries in trade; when they are hit by financial shocks, their exporter clients sell less abroad. This paper connects international trade and banking crises through a concrete firm-bank channel.